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Fire Constrains Orchid Occurrence in Chapada Diamantina, Brazil

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ABSTRACT

Fires are key ecological disturbances in tropical ecosystems and are shifting in response to human activities, underscoring the need to understand species-specific responses. Their impacts on orchids remain poorly understood, particularly in biodiverse tropical mosaics such as the Chapada Diamantina region in Brazil. Here, we investigate the relationship between the occurrence of endemic orchids and the time since the last fire (TSLF), testing whether responses are positive, negative, or mixed. We applied a Bayesian hierarchical model within a use-availability framework to analyse records of endemic orchids sampled across a 38-year fire chronosequence, both for orchids at a group level and for individual species. As a group, orchids showed a positive association with TSLF, indicating that orchids are generally suppressed following fire. Individual species exhibited little variation in their responses, with no significant species-specific effects detected. Nevertheless, our results suggest that orchids are generally sensitive to fire, likely because of ecological mechanisms that warrant further investigation. These findings highlight that fire represents a significant threat to endemic orchids in the Chapada Diamantina, with important implications for their conservation and for fire management in this Brazilian biodiversity hotspot.

1 | Introduction

Fire poses significant environmental and socioeconomic challenges when driven by inadequate crop and livestock management. Such practices, combined with climate change, are driving global shifts in fire regimes, particularly in tropical ecosystems (Hantson et al. 2024). Notably, tropical ecosystems exhibit varying responses to burning, ranging from fire-adapted to fire-sensitive ecosystems and species (Schmidt et al. 2018). Savanna-forest mosaics may respond to fire regimes in complex ways, with different species being positively or negatively affected (Brundrett 2025).

This global trend is mirrored in Brazil, which records over 300 000 fires annually across all biomes, particularly during the

dry season, primarily because of inadequate management practices (Costa et al. 2023). The Chapada Diamantina region, in Bahia State, is annually affected by fires. It harbours the Chapada Diamantina National Park, one of the most affected Brazilian National Parks by fires per season (Gonçalves et al. 2011; Mesquita et al. 2011). Fires in the region occur mainly during the dry season (August–November), affecting most vegetation types, ranging from savanna to forest formations (Brasil 2007). Approximately 68% of this protected area has been burned at least once, with fire frequency, intensity, and time since the last fire (TSLF) being highly variable and influenced by multiple factors, such as landscape heterogeneity, land use, distance to the nearest municipality, tree cover, and altitude (Anjos et al. 2022). This region is also an important centre of biodiversity and endemism owing to its topographical heterogeneity and intermingling

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of the Atlantic Forest, Cerrado, and Caatinga (Azevedo and Van Den Berg 2007).

Orchids (Asparagales: Orchidaceae) illustrate the high biodiversity and endemism of the Chapada Diamantina region, with 11 genera and 16 endemic species distributed across various vegetation types, including deciduous forests, grasslands, and shrublands, and occupying different substrates, such as rupicolous, epiphytic, and tericolous substrates (Van Den Berg and de Azevedo 2005; Azevedo and Van Den Berg 2007; Gonçalves and de Gonçalves 2023). Orchids play a central role in ecosystems by providing resources for many native insects and have high economic value in the floriculture and pharmaceutical industries (Tiwari et al. 2024). Furthermore, orchids are indicators of environmental resilience owing to their diversity, interaction networks with microbiota and insects, and survival strategies (Goswami et al. 2024). However, the responses of orchids to burning vary. Although fire regimes may favour many species through increased flowering and pollination, others are negatively affected or prefer long-unburned areas (Brundrett 2025). These contrasting responses highlight the need for a careful examination of orchid responses to fire. This issue remains poorly understood in Brazilian ecosystems, including well-studied fire-adapted ecosystems such as the Brazilian Cerrado. Recent studies have summarised the varied effects of fire on plants (Fidelis and Zironi 2021) and the lack of effect on plant-pollinator networks in the Cerrado ecoregion (Baronio et al. 2021). However, they provided no clear evidence of positive or negative effects on orchids. To our knowledge, no study has addressed the effects of TSLF on orchids in Brazil.

The unique biogeographical and ecological context of the Chapada Diamantina region, with high levels of endemism and fire incidence, makes it essential to study the response of orchids to fires. This investigation may reveal how this disturbance

drives population dynamics, plant-pollinator networks, and tropical plant communities. It is also crucial to tailor conservation measures for protected areas and assess the economic impact of fires on native orchids in the wild. We first assessed this question by combining remote sensing and species occurrence data using a Bayesian hierarchical presence-availability modelling approach. We investigated whether the occurrence records of endemic orchids in the Chapada Diamantina region were positively or negatively related to the TSLF.

2 | Methods

The study area comprised the Chapada Diamantina region, located in the northern portion of the Espinhaço mountain chain. We delimited the study area as the extent of occurrence of orchid species considered endemic to this region (Azevedo and Van Den Berg 2007; Gonçalves and de Gonçalves 2023). It includes Chapada Diamantina National Park, all the municipalities where the protected area intersects, and their adjacencies (Figure 1). The region is situated in an ecotone between the Caatinga, Cerrado, and Atlantic Rainforest formations, and the predominant vegetation includes rocky outcrops, savannas, deciduous forests, and evergreen forests. Its local climate is mesothermic, with average temperatures of approximately 22°C and a rainy season between November and March (Juncá 2005). The region experiences fires annually of varying extents and intensities, from both natural and anthropogenic causes. Despite the limited knowledge of fire regimes in the region, studies have found that vegetation recovers rapidly from fires (Gonçalves et al. 2011).

Ideally, the effect of fires on plant occurrence requires assessment of species' occurrence before and after fires. However,

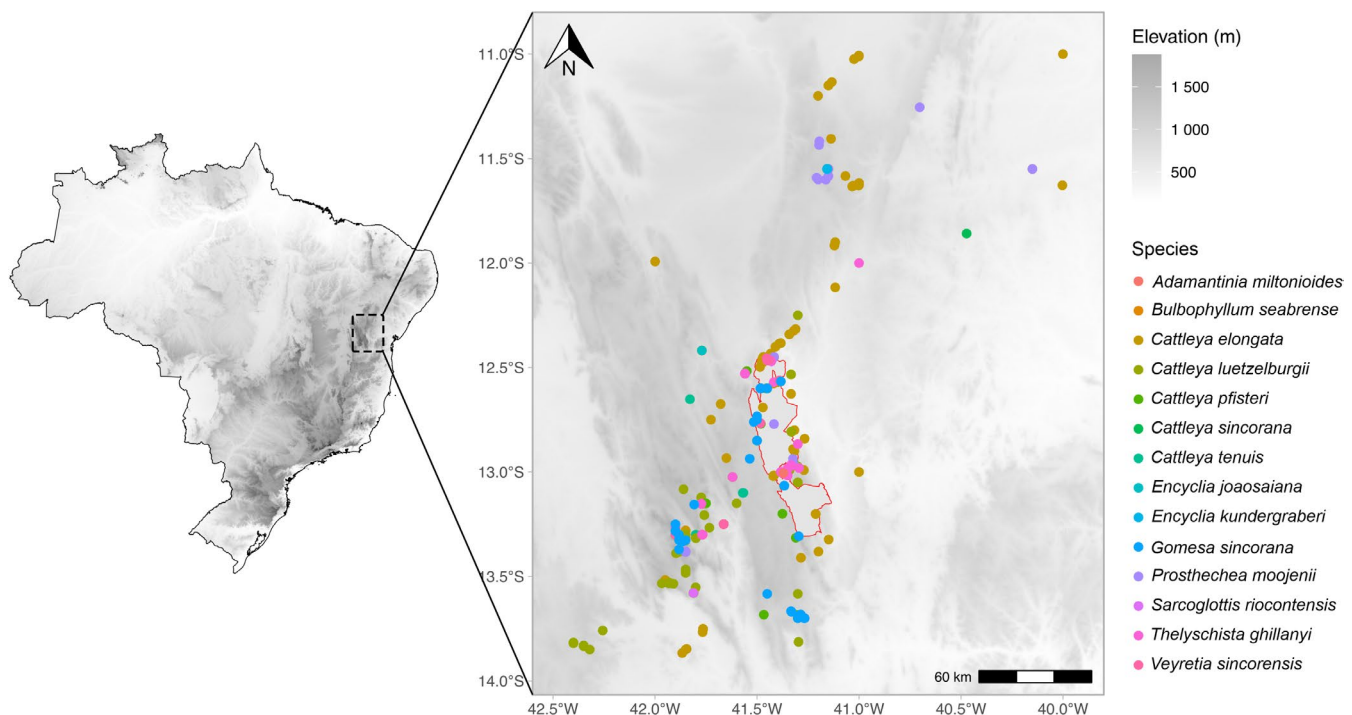


FIGURE 1 | Location of the study area, Chapada Diamantina region in Brazil, and occurrences of endemic orchids. The red contour highlights the Chapada Diamantina National Park.

long-term monitoring of plant populations is lacking in most Brazilian ecosystems, particularly in relation to the fire regimes (Lucas et al. 2023). Therefore, we investigated potential relationships between orchid records and fire events in the study region through a 38-year recording of fire events.

We searched for occurrence records (i.e., geographic coordinates) of endemic orchids and the year of each record using the GBIF (2024) and speciesLink (www.specieslink.net) databases. We also gathered information on fire incidence in the study area from 1985 to 2022 from MapBiomas (2024). For each occurrence record, we drew a circular buffer area with a radius of 50 m (Wilkin et al. 2021), from which we determined the time since the last fire (TSLF) at each location.

First, we explored whether both species records and fire occurrence followed any particular temporal trend that could compromise any inference. For example, if either species records or fire incidence increased with time, then we could, by chance, find systematically shorter times since the last fire before the species record was associated with later records. This effect would bias our inferences. After ensuring no significant temporal trends in either orchid records or fire occurrence (Figure 2), we employed a use-availability model to test the relationship between species occurrence and TSLF (Johnson et al. 2021).

The model requires defining a set of pseudo-absences distributed across the study area. To draw pseudo-absences, we defined a buffer of 200 km around the occurrence records of the species. For each species, we generated pseudo-absences that were 10 times greater than their presence. We generated environmentally stratified pseudo-absences within the buffer area

to avoid selecting ecologically unrealistic locations while enforcing a minimum distance of 100 m from any presence point to reduce spatial dependence. Pseudo-absences were restricted to the land-cover classes and elevation ranges observed at occurrence points, based on the land use and cover provided by MapBiomas Col. 9 (MapBiomas 2024). We reclassified the land cover into four classes: forest, grassland, agriculture, and water bodies. The elevation strata were defined using the quantiles of elevation measured at the occurrence locations. Elevation data were obtained from the INPE (Valeriano 2004). We then combined the presence and pseudo-absence data for subsequent analysis.

We fitted a Bayesian hierarchical model using the ‘brms’ package (Bürkner 2024) in the R environment version 4.4.2 (R Core Team 2024). The model was implemented using Hamiltonian Monte Carlo sampling, with the response variable modelled as a Bernoulli distribution with a logit link function given as follows:

$$\text{logit}(P(\text{occ} = 1)) = \beta_0 + \beta_1 \Delta t + (1|sp)$$

where β_0 is the intercept, β_1 is the effect of time since the last fire (Δt), and the random term $(1|sp)$ provides the intercept specific to each species.

This model enabled us to determine the overall effect on all orchid species as a group and the individual effect on each species. Species with few records are unable to have potentially significant responses detected. Despite the issue of sampling heterogeneity among species, which increases the type II error rate, fewer occurrences of these less-sampled species help us draw the overall effect for orchids as a group. Individual species responses can only be inferred from those with a large sample size for the model.

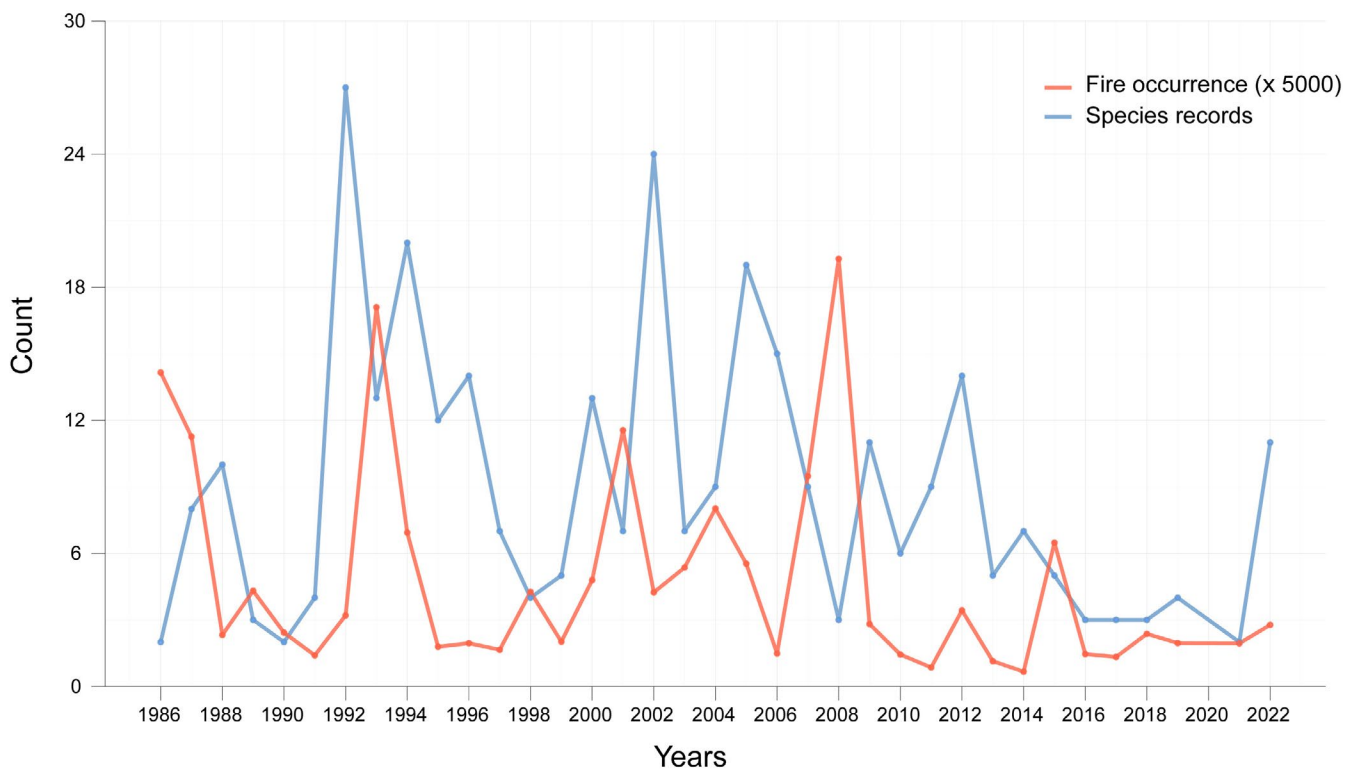


FIGURE 2 | Temporal trends in the numbers of orchid records and fire occurrence in the study region over 38 years of fire incidence records in the Chapada Diamantina region in Brazil.

We specified weakly informative priors to improve model stability and prevent unrealistic estimates, with a normally distributed intercept (0,2) and fixed effect (0,1), and random effect standard deviation with a Cauchy distribution (0,1). We used four chains with 8000 iterations each (4000 warm-up iterations) to ensure proper convergence and address sampling issues. We set the step size adaptation target (adapt_delta) to 0.99 to reduce divergent transitions and the maximum depth of the binary tree (max_treedepth) to 20 to allow deeper parameter space exploration. Model convergence was ensured with Rhat statistics close to 1.0, indicating the good mixing of the Markov chains.

When the relationship is negative, the probability of a species record is higher at shorter times since the last fire, indicating a positive effect of fire on species occurrence (as inferred from the records). Conversely, a positive relationship indicates that records are more likely to be found after longer periods since the last fire, suggesting that the species is negatively affected by fire.

3 | Results

We found 381 records for 14 species, *Cattleya elongata* (166), *Cattleya luetzelburgii* (72), *Thelyschista ghillanyi* (41), *Gomesa sincorana* (30), *Prosthechea moojenii* (23), *Cattleya pfisteri* (13), *Cattleya tenuis* (10), *Veyretia sincorensis* (10), *Cattleya sincorana* (8), *Encyclia kundergraberii* (4), *Adamantinia miltonioides* (1), *Bulbophyllum seabrense* (1), *Encyclia joaosaiana* (1) and *Sarcoglottis riocontensis* (1) (Figure 1). The time since the last fire (TSLF) ranged from one to 38 years. Fire events and species records varied haphazardly in incidence throughout the years in the study region, suggesting that the two are independent of each other (Figure 2).

The Bayesian hierarchical model showed a positive relationship between species occurrence and TSLF when all species were pooled. The fixed effect had an estimated intercept of -3.34 (95% CI: -4.20 to -2.70), indicating a very low probability (3.34%) of occurrence immediately after a fire. The model also had a standardised coefficient of 0.05 (95% CI: 0.03–0.07), indicating an average annual increase of 5% in the probability of records after fire incidence (Figure 3). Taken together, these results corroborate a negative effect of fire on species detection. The model achieved good convergence, with Rhat=1.00 for all parameters and high effective sample sizes (Bulk ESS=814 and Tail ESS=631), ensuring reliable posterior estimates (Table S1–S2).

The random intercepts for species had a standard deviation of 0.59 (95% CI: 0.01–1.58), indicating moderate variability in species-specific occurrence probabilities. Random slopes for the effect of TSLF were minimal (SD=0.01, 95% CI: 0.00–0.04), suggesting that species respond similarly to the time since the last fire. The correlation between species intercepts and slopes was negative (-0.79 , 95% CI: -1.00 to 0.54), although there was wide uncertainty (Table S1–S2).

Regarding species-specific effects, no species exhibited a statistically significant response to TSLF, either positively or negatively (Table S2). The intercepts varied among species, reflecting differences in baseline occurrence probabilities. All species had slope estimates close to zero, with 95% credible intervals largely overlapping (e.g., *Cattleya elongata*: 0.016, 95% CI: -0.002 to 0.041; *Gomesa sincorana*: -0.012 , 95% CI: -0.041 to 0.007), confirming that the positive effect of the TSLF was consistent across species. Detailed estimates of the fixed effects, hyperparameters, and species-specific random effects (intercepts and slopes) are provided in Tables S1–S2 and S2.

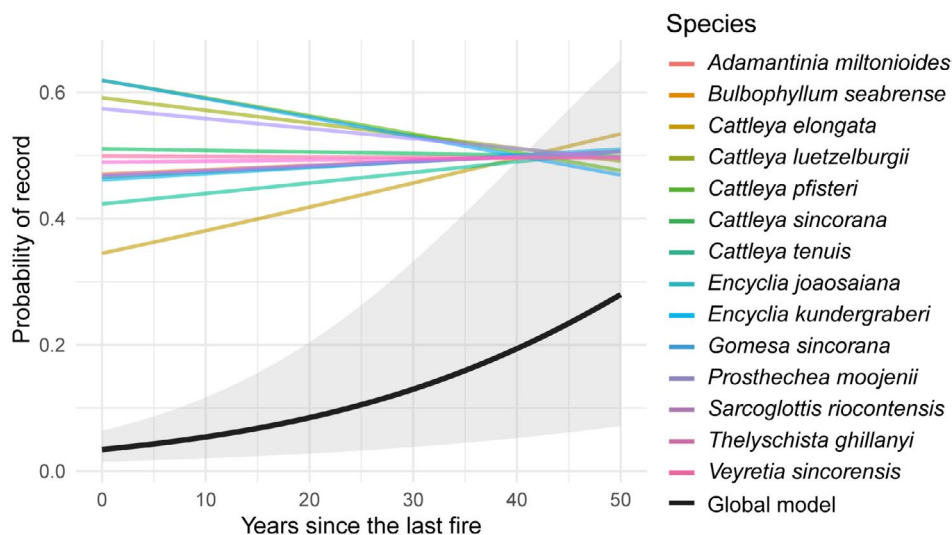


FIGURE 3 | Relationships between the predicted probability of occurrence and the time since the last fire for all species together (black) and for each species (colours) of endemic orchids from the Chapada Diamantina region based on a Bayesian hierarchical model. Positive relationships indicate a higher probability of occurrence with a longer time since the last fire, denoting sensitivity to fire. Negative relationships indicate a higher probability of occurrence at shorter times since the last fire, suggesting a fire-stimulated occurrence. The global intercept appears lower than all individual intercepts because, in multilevel models, the global curve reflects only the mean fixed effect, whereas the individual species curves include their specific random deviations.

4 | Discussion

Our results indicate that the time since the last fire (TSLF) affects orchid occurrence across the studied species. The global effect of TSLF was positive, indicating that longer periods without fire were associated with a higher probability of occurrence, thus suggesting sensitivity to fire (Figure 3). Species-specific responses were broadly similar, as indicated by the minimal random slopes, although baseline occurrence probabilities (intercepts) varied moderately among species. These findings suggest that fire acts as a suppressive factor, reducing the likelihood of orchid occurrence following burning, whereas longer fire-free intervals allow populations to establish. Although we found no evidence of species-specific sensitivity to fire, our findings support the fire sensitivity of orchids as a group.

Few studies have examined the effects of fire on plants in the Chapada Diamantina region. Although no clear relationship has been established between species composition and fire recurrence, some plant groups appear to benefit from fire, whereas others are negatively affected (Gonçalves et al. 2011). Evidence suggests an overall reduction in the establishment of species regenerated from seeds, accompanied by a higher prevalence of species resprouting from perennial organs after fire events (e.g., Neves and Conceição 2010). This pattern is particularly evident in graminoid clades such as the pipeworts (Eriocaulaceae). Local species like *Comanthera mucugensis* (Giul.) L.R. Parra & Giul., which are economically important in the region, have traditionally been managed through burning to stimulate regrowth (Gonçalves et al. 2011).

Orchid assemblages may also respond to fire positively, negatively, or in both ways (Brundrett 2025). The constraint posed by fire on orchids in the region may be related to the local fire regime, which is related to climatic conditions and habitat types. The effects of fire on orchids may be modulated by climate (especially rainfall) and habitat type (Bell et al. 2021). Accordingly, burning may increase flowering and population size under wet conditions, but may be detrimental during drier seasons. In addition, forest and open habitats may respond differently, with forests often showing greater resilience (Jasinge et al. 2018). In fact, the period with a higher risk of fire in the region is the dry season, which occurs between August and November (Brasil 2007). In addition, most of the Chapada Diamantina region is covered by savannas (i.e., Cerrado), shrublands (i.e., Caatinga), and mosaics of these formations, together with deciduous forests (Funch et al. 2009). These conditions likely exacerbate the negative effects of fire on species in the region. Increases in frequency or severity of fire of anthropic origin may compromise recruitment and recovery of populations, increasing the risk of extinction (Kelly et al. 2020).

The mechanisms involved in the orchid's susceptibility to fire may vary. They may entail different combinations of burning properties, species traits, and community assembly dynamics (Pellegrini et al. 2023). Burning properties, such as fire intensity and duration, may have varying consequences for nutrient availability and community structure, thereby affecting species responses to fire (Brundrett 2025). In turn, species traits such as substrate, tuber presence and depth, and seed requirements for dormancy break may reflect a species' sensitivity to fire (Duncan 2012). Most

species included in our analyses (11 of 14, ~80%) are rupicolous or epiphytic (do Brasil 2025), which typically lack tubers, making them more susceptible to fire (Duncan 2012). Finally, the susceptibility may also be linked to post-fire vegetation recovery and succession, which may favour the increased recruitment of fire-tolerant species while reducing other less tolerant species, such as orchids. This pattern appears to be the case in plant communities from the Cerrado ecoregion, where fire frequency promotes alterations in species occurrence and abundance, thereby changing the local plant community (Rodrigues and Fidelis 2022). Overall, the existence of a group-level response to fire, combined with a lack of detailed data and clarification of the mechanisms involved, requires further field studies and trait-based approaches to assess the impact of fire on the survivorship, recruitment, and population structure of these species (Kelly et al. 2020).

Although our findings were restricted to clarifying the mechanism underlying orchids' susceptibility to fire, they demonstrate the pattern, thereby providing strong evidence against fire incidence in the Chapada Diamantina region, particularly in the National Park. According to the management plan for the park (Brasil 2007), fire is the greatest threat in the region, along with mining and agriculture. The Park's strategy for fighting and preventing fires includes public awareness campaigns and the maintenance of a community-based fire brigade. Leveraging these measures to prevent fires should reduce the risk of population loss and eventual extinction, ensuring the protection of the orchids' high diversity and endemism in the region.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available in GBIF at <http://www.gbif.org>. These data were derived from the following resources available in the public domain: Species Link, <http://www.specielink.net>; Mapbiomas, <https://doi.org/10.58053/MapBiomas/XUFVIC>.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Tables S1–S2:** aec70160-sup-0001-supinfo.docx.